torch.copysign#

https://pytorch.org/docs/stable/generated/torch.copysign.html#torch.copysign

Description:

Create a new floating-point tensor with the magnitude of input and the sign of other, elementwise. Supports broadcasting to a common shape, and integer and float inputs. input (Tensor) – magnitudes. other (Tensor or Number) – contains value(s) whose signbit(s) are applied to the magnitudes in input. out (Tensor, optional) – the output tensor.

Function Signature

```
torch.copysign(input, other, *, out=None) → Tensor#
Parameters
Keyword Arguments
```

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.randn(5)
>>> a
tensor([-1.2557, -0.0026, -0.5387,  0.4740, -0.9244])
>>> torch.copysign(a, 1)
tensor([1.2557, 0.0026, 0.5387, 0.4740, 0.9244])
>>> a = torch.randn(4, 4)
>>> a
tensor([[ 0.7079,  0.2778, -1.0249,  0.5719],
        [-0.0059, -0.2600, -0.4475, -1.3948],
        [ 0.3667, -0.9567, -2.5757, -0.1751],
        [ 0.2046, -0.0742,  0.2998, -0.1054]])
>>> b = torch.randn(4)
tensor([ 0.2373,  0.3120,  0.3190, -1.1128])
>>> torch.copysign(a, b)
tensor([[ 0.7079,  0.2778,  1.0249, -0.5719],
        [ 0.0059,  0.2600,  0.4475, -1.3948],
        [ 0.3667,  0.9567,  2.5757, -0.1751],
        [ 0.2046,  0.0742,  0.2998, -0.1054]])
>>> a = torch.tensor([1.])
>>> b = torch.tensor([-0.])
>>> torch.copysign(a, b)
tensor([-1.])
```

Notes & Warnings

> **NOTE** Note copysign handles signed zeros. If the other argument has a negative zero (-0), the corresponding output value will be negative.

Detailed Documentation

Documentation

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Supports broadcasting to a common shape, and integer and float inputs.

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